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Si dos funciones diferenciables coinciden en un entorno de p, sus derivadas coinciden en p

Lema 1. *Sea $p \in M$ variedad diferenciable y $f, g \in \mathcal{C}^\infty(M)$, entonces*

$$\exists U \in \mathcal{V}(p) : f\big|_U = g\big|_U \implies \forall w \in T_pM : w(f) = w(g).$$

***Demostración*:** Sea $h = f - g$ (entonces $h\big|_U = 0$). Sea $\psi \colon M \longrightarrow \mathbb{R}$ una función bump para $\operatorname{supp}(h)$ con soporte en $U \setminus \{p\}$. Es decir, $\psi(\operatorname{supp}(h)) = 1 \wedge \psi(p) = 0$. Observamos

(i) $\psi \cdot h = h$ porque $\forall q \in M : h(q) = 0 \vee (h(q) \neq 0 \implies q \in \operatorname{supp}(h) \implies \psi(q) = 1)$.

(ii) $\psi(p) = 0 \wedge h(p) = 0$ por lo que $w(\psi h) = 0$. Usando el apartado anterior tenemos $w(\psi h) = w(h) = w(f - g) = w(f) - w(g) = 0$.

Por tanto concluimos que $\forall w \in T_pM : w(f) = w(g)$. ■

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